ODE HW CH 8

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problem 1

predator prey system

$$F' = 0.2F - 0.1FS, \tag{1}$$
$$S' = -0.3S + 0.1FS, \tag{2}$$

where $F(t)$ is the prey population and $S(t)$ is the predator population.

(a) phase plane analysis

all trajectories form **closed orbits** surrounding the equilibrium point. as F increases the predator population rises; as S increases the prey is driven down, as prey declines the predators starve and decrease; and as predators decrease the prey recovers, completing one full cycle. the populations oscillate infinitely without tending toward either extinction or high growth.

(b) solution starting at $F(0) = 3$, $S(0) = 2$

the equilibrium of system (1)–(2) is found by setting $F' = S' = 0$:

$$F^* = \frac{\gamma}{\delta} = \frac{0.3}{0.1} = 3, \qquad S^* = \frac{\alpha}{\beta} = \frac{0.2}{0.1} = 2.$$

the initial condition $F(0) = 3$, $S(0) = 2$ fits exactly with this equilibrium. therefore the solution is **constant for all time**:

$$F(t) \equiv 3, \qquad S(t) \equiv 2, \qquad t \geq 0.$$

both populations remain at their equilibrium values indefinitely.

problem 2

$$x' = 0.2x - 0.04xy, \tag{3}$$
$$y' = -0.1y + 0.005xy. \tag{4}$$

(i) nullclines

x**-nullclines** (set $x' = 0$):

$$0.2x - 0.04xy = x(0.2 - 0.04y) = 0 \implies x = 0 \quad \text{or} \quad y = 5.$$

y**-nullclines** (set $y' = 0$):

$$-0.1y + 0.005xy = y(-0.1 + 0.005x) = 0 \implies y = 0 \quad \text{or} \quad x = 20.$$

the x-nullclines ($x = 0$ and $y = 5$) and y-nullclines ($y = 0$ and $x = 20$) are plotted in the phase plane below

Uncomment once nullclines.png is in the same folder as this .tex file.

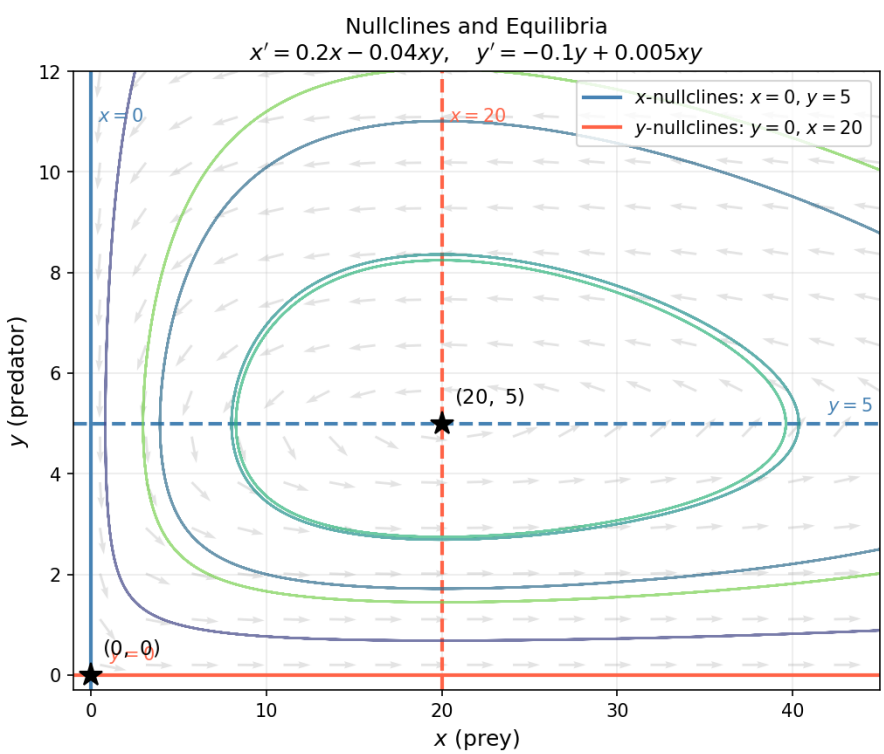

Figure 1: nullclines and equilibrium for system (3)–(4). blue lines are x-nullclines; red lines are y-nullclines. black stars mark the equilibrium points $(0, 0)$ and $(20, 5)$.

(ii) equilibrium points

when an x-nullcline intersects a y-nullcline, where $x' = 0$ and $y' = 0$ simultaneously.

x-nullcline	y-nullcline	equilibrium	
$x = 0$	$y = 0$	$(0,\ 0)$	
$x = 0$	$x = 20$	$(0,\ 0)$	(same point)
$y = 5$	$y = 0$	$(0,\ 0)$	(same point)
$y = 5$	$x = 20$	$(20,\ 5)$	

There are therefore **two distinct equilibrium points**:

$$(x^*, y^*) = (0,\ 0) \qquad \text{and} \qquad (x^*, y^*) = (20,\ 5).$$

problem 3

consider the system

$$x_1' = 8x_1 - 10x_2, \tag{5}$$
$$x_2' = 5x_1 - 7x_2. \tag{6}$$

(i) matrix–vector form

system (5)–(6) can be written as $\mathbf{x}' = A\mathbf{x}$, where

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad A = \begin{pmatrix} 8 & -10 \\ 5 & -7 \end{pmatrix}.$$

which is,

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 8 & -10 \\ 5 & -7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

(ii) verify that $\mathbf{x}(t) = (2e^{3t},\ e^{3t})^\mathsf{T}$ is a solution

let

$$\mathbf{x}(t) = \begin{pmatrix} 2e^{3t} \\ e^{3t} \end{pmatrix}.$$

verify that $\mathbf{x}' = A\mathbf{x}$ by confirming they are equal.

left side — differentiate $\mathbf{x}$

$$\mathbf{x}'(t) = \begin{pmatrix} \dfrac{d}{dt}(2e^{3t}) \\ \dfrac{d}{dt}(e^{3t}) \end{pmatrix} = \begin{pmatrix} 6e^{3t} \\ 3e^{3t} \end{pmatrix}.$$

right side — multiply $A\mathbf{x}$:

$$A\mathbf{x} = \begin{pmatrix} 8 & -10 \\ 5 & -7 \end{pmatrix} \begin{pmatrix} 2e^{3t} \\ e^{3t} \end{pmatrix} = \begin{pmatrix} 8(2e^{3t}) - 10(e^{3t}) \\ 5(2e^{3t}) - 7(e^{3t}) \end{pmatrix} = \begin{pmatrix} 16e^{3t} - 10e^{3t} \\ 10e^{3t} - 7e^{3t} \end{pmatrix} = \begin{pmatrix} 6e^{3t} \\ 3e^{3t} \end{pmatrix}.$$

since $\mathbf{x}'(t) = A\mathbf{x}(t)$ for all t, the vector-valued function $\mathbf{x}(t) = (2e^{3t}, e^{3t})^{\mathsf{T}}$ is a solution of the system.